MIL-M-0038510/2F(USAF)
24 OCT 1975
USED IN LIEU OF
MIL-M-38510/2E
24 December 1974

MILITARY SPECIFICATION

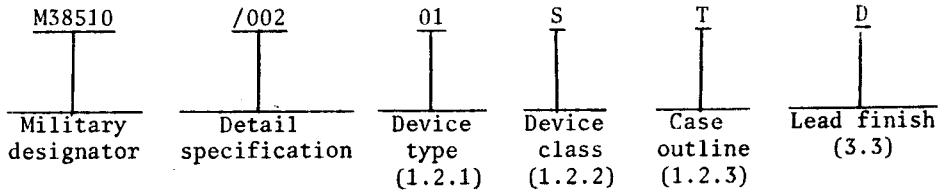
MICROCIRCUITS, DIGITAL, TTL, FLIP-FLOPS MONOLITHIC SILICON

This limited coordination Military specification has been prepared by the Air Force Space and Missile Systems Organization based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of Military Specification MIL-M-38510/2E. It is subject to modification. However, pending its promulgation as a coordinated Military specification, it may be used in procurement.

1. SCOPE

1.1 Scope. This specification covers the requirements for monolithic silicon, TTL, bistable logic microcircuits for use in Air Force missile and space systems.

1.2 Part number. The complete part number shall be as shown in the following example:



1.2.1 Device type. The device type shall be as shown in the following:

Device type	Circuit	Similar commercial type
01	Single J-K master-slave flip-flop	5472T
02	Dual J-K master-slave flip-flop, no preset	5473T
05	Dual D-type edge-triggered flip-flop	5474T

1.2.2 Device class. The device class shall be class S as defined in MIL-M-0038510.

1.2.3 Case outline. The case outline shall be as specified in figure 10 of MIL-M-0038510 and shall be identified as outline T.

1.3 Absolute maximum ratings.

Supply voltage range	0.5 to 8.0 Vdc
Input voltage range	0 Vdc to 5.5 Vdc
Storage temperature range	-65°C to 150°C
Lead temperature (soldering, 10 seconds)	300°C

1.4 Recommended operating conditions.

Ambient operating temperature range	-55°C to 125°C
Supply voltage	4.5 Vdc minimum to 5.5 Vdc maximum
Minimum high level input voltage	2.0 Vdc
Maximum low level input voltage	0.8 Vdc

FSC 5962

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2. APPLICABLE DOCUMENTS

2.1 The following document, of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-0038510 - Microcircuits, General Specification for.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-0038510 and as specified herein.

3.2 Design, construction, weight and physical dimensions. The design, construction, weight and physical dimensions shall be as specified in MIL-M-0038510 and herein.

3.2.1 Logic diagram and terminal connections. The logic diagram and terminal connections shall be as specified on figure 1.

3.2.2 Truth tables and logic equations. The truth tables and logic equations shall as specified on figure 2.

3.2.3 Schematic circuit. The schematic circuit shall be as specified on figure 3.

3.2.4 Case outline. The case outline shall be as specified in 1.2.3.

3.2.5 Weight. The maximum weight shall be 1 gram.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-0038510.

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in tables I and II.

3.4.1 Delta drift limits. The delta drift limits shall be as specified in table I.

3.5 Screening. Screening shall be in accordance with table I of MIL-M-0038510 and as specified herein.

3.5.1 Interim electrical parameters. The interim electrical parameters shall consist of all the electrical characteristics tests as specified in table I. Each test shall be performed go, no-go at -55°C and 125°C. Each test shall also be read and recorded at 25°C. For all tests, the devices shall meet the limits specified in table I.

3.5.2 Final electrical tests. The final electrical tests shall consist of all the electrical characteristics tests specified in table I. Each test shall be read and recorded at -55°C, 25°C and 125°C. For all tests, the devices shall meet the limits specified in table I.

3.5.2.1 Delta computation. Delta computations shall be performed for those parameters in table I which are marked with an asterisk. All devices shall meet the applicable delta limits specified in table I. (See section 6 in MIL-M-0038510.)

3.5.3 Switching characteristics tests. The switching characteristics tests specified in table II and figures 4 through 9 shall be performed at 25°C. All devices shall meet the limits specified in table II.

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TABLE I. (Device type 01) Electrical characteristics tests. 4/5/

1/2/ TEST NUMBER	TEST NAME	PIN NUMBERS 6/														3/ LIMITS		1/ DELTA Δ	UNITS
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	MIN	MAX		
1*	V_{OH}	0.8 V	A		4.5 V			2.0 V	2.0 V	2.0 V		GND	-.4 mA	0.8 V	0.8 V	2.4		±.120	V
2*	V_{OH}	2.0 V	A					0.8 V	0.8 V	0.8 V	-.4 mA			2.0 V	2.0 V				
3*	V_{OH}			0.8 V		2.0 V							-.4 mA						
4*	V_{OH}			2.0 V		0.8 V					-.4 mA								
5*	V_{OL}	2.0 V	A					0.8 V	0.8 V	0.8 V			16 mA	2.0 V	2.0 V		0.4	±.04	
6*	V_{OL}	0.8 V	A					2.0 V	2.0 V	2.0 V	16 mA			0.8 V	0.8 V				
7*	V_{OL}			0.8 V		2.0 V					16 mA								
8*	V_{OL}			2.0 V		0.8 V							16 mA						
9	V_{IC}							-12 mA									-1.5		
10									-12 mA										
11										-12 mA									
12		-12 mA																	
13														-12 mA					
14															-12 mA				
15						-12 mA													
16				-12 mA															
17			-12 mA																
18	I_{IL1}	GND	4.5 V		5.5 V	B		0.4 V	4.5 V	4.5 V				GND	GND	-0.7	-1.6		mA
19						B		4.5 V	0.4 V	4.5 V				GND	GND				
20						B		4.5 V	4.5 V	0.4 V				GND	GND				
21		0.4 V		B				GND	GND	GND				4.5 V	4.5 V				
22		4.5 V		B				GND	GND	GND				0.4 V	4.5 V				
23		4.5 V		B				GND	GND	GND				4.5 V	0.4 V				
24	I_{IL2}	4.5 V	0.4 V	B				4.5 V	4.5 V	4.5 V				4.5 V	4.5 V	-1.4	-3.2		
25	I_{IL2}		0.4 V			B								4.5 V	4.5 V	-1.4	-3.2		
26	I_{IL3}			0.4 V										4.5 V	4.5 V	-0.7	-3.2		
27	I_{IL3}					0.4 V								4.5 V	4.5 V	-0.7	-3.2		
28*	I_{IH1}	GND			5.5 V	GND		2.4 V	GND	GND						0	40	10X or 4 μA	μA

See notes at end of device type 01.

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TABLE I. (Device type 01) Electrical characteristics tests. 4/5/ - Continued

1/2/ TEST NUMBER	TEST NAME	PIN NUMBERS 6/														3/ LIMITS		1/ DELTA Δ	UNITS
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	MIN	MAX		
29*	I_{IH1}		GND		5.5 V	GND		GND	2.4 V	GND		GND				0	40	10X or 4 µA	µA
30*						GND		GND	GND	2.4 V									
31*		2.4 V		GND										GND	GND				
32*		GND		GND										2.4 V	GND				
33*		GND		GND										GND	2.4 V				
34	I_{IH2}					GND		5.5 V	GND	GND							100		
35						GND		GND	5.5 V	GND									
36						GND		GND	GND	5.5 V									
37		5.5 V		GND										GND	GND				
38		GND		GND										5.5 V	GND				
39		GND		GND										GND	5.5 V				
40	I_{IH3}					2.4 V		GND	GND	GND							80		
41		GND		2.4 V										GND	GND				
42			2.4 V			GND		GND	GND	GND				GND	GND				
43			2.4 V	GND				GND	GND	GND				GND	GND				
44	I_{IH4}		GND	5.5 V										GND	GND		200		
45			GND			5.5 V		GND	GND	GND									
46			5.5 V			GND								GND	GND				
47			5.5 V	GND										GND	GND				
48	I_{OS}	4.5 V	GND	GND				4.5 V	4.5 V	4.5 V			GND	4.5 V	4.5 V	-20	-57		mA
49	I_{OS}	4.5 V				GND		4.5 V	4.5 V	4.5 V	GND			4.5 V	4.5 V	-20	-57		
50	I_{CC}			GND													20		
51	I_{CC}					GND											20		

1/ Delta computation shall be performed on parameters marked with an asterisk (*) with data read and recorded ("10X" denotes "10 times the initial measurement"). Where two delta values are indicated for one parameter, the greater value shall apply.
2/ Test sequence is optional with the manufacturer.
3/ Unless otherwise specified, the limits apply at T_A = -55°C to +125°C.
4/ A = nominal clock pulse; B = momentary GND, then 4.5 V.
5/ V_{IC} tests apply at 25°C only.
6/ Unused input pins may be connected to 4.5 V.

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TABLE I. (Device type 02) Electrical characteristics tests. 4/5/

1/2/ TEST NUMBER	TEST NAME	PIN NUMBERS 6/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	3/ LIMITS MIN	MAX	1/ DELTA Δ	UNITS
1*	V_{OH}	A		0.8 V	4.5 V							GND	-.4 mA		2.0 V	2.4		±.120	V
2*		A		2.0 V										-.4 mA	0.8 V				
3*			0.8 V											-.4 mA					
4*						A		2.0 V		-.4 mA	0.8 V								
5*						A		0.8 V	-.4 mA		2.0 V								
6*							0.8 V		-.4 mA										
7*	V_{OL}	A		2.0 V									16 mA		0.8 V		0.4	±.04	
8*		A		0.8 V										16 mA	2.0 V				
9*			0.8 V										16 mA						
10*						A		0.8 V		16 mA	2.0 V								
11*						A		2.0 V	16 mA		0.8 V								
12*							0.8 V			16 mA									
13*	V_{IC}														-12 mA		1.5		
14								-12 mA											
15				-12 mA															
16											-12 mA								
17			-12 mA																
18		-12 mA																	
19							-12 mA												
20						-12 mA													
21*	I_{IL1}	4.5 V	4.5 V		5.5 V								C		0.4 V	-0.7	-1.6	±.02	mA
22*		4.5 V	4.5 V	0.4 V										C					
23*						4.5 V	4.5 V	0.4 V		C									
24*						4.5 V	4.5 V		C		00.4 V								
25	I_{IL2}	0.4 V	B	4.5 V											4.5 V	-1.4	-3.2		
26				4.5 V		0.4 V	B	4.5 V			4.5 V					-1.4	-3.2		
27	I_{IL3}	4.5 V	0.4 V												4.5 V	-0.7	-1.6		
28						4.5 V	0.4 V	4.5 V			4.5 V					-0.7	-1.6		

See notes at end of device type 02.

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TABLE I. (Device type 02) Electrical characteristics tests. 4/5/ - Continued

1/2/ TEST NUMBER	TEST NAME	PIN NUMBERS 6/														3/ LIMITS		1/ DELTA Δ	UNITS
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	MIN	MAX		
29*	I_{IH1}	GND	GND		5.5 V							GND			2.4 V	0	40	10X or 4 µA	µA
30*		GND	B	2.4 V															
31*						GND	GND	2.4 V											
32*						GND	B				2.4 V								
33	I_{IH2}	GND	GND												4.5 V		100		
34		GND	B	5.5 V															
35						GND	GND	5.5 V											
36						GND	B				5.5 V								
37*	I_{IH3}	GND	2.4 V											C	GND		80	10X or 8 µA	
38*	I_{IH3}					GND	2.4 V	GND	C								80		
39*	I_{IH4}	5.5 V	GND	GND											GND		200		
40		GND	5.5 V										C		GND				
41						5.5 V	GND	GND			GND								
42						GND	5.5 V	GND	C										
43*	I_{IH5}	2.4 V	GND	GND											GND	-850	80	10X or 8 µA	
44*	I_{IH5}					2.4 V	GND	GND			GND					-850	80		
45	I_{OS}	2.4 V	GND	2.4 V										GND	2.4 V	-20	-57		mA
46 5/		GND	2.4 V	2.4 V									GND	GND	2.4 V				
47 5/						GND	2.4 V	2.4 V	GND	GND	2.4 V								
48						2.4 V	GND	2.4 V	GND		2.4 V								
49	I_{CC}	D	4.5 V	GND		D	4.5 V	4.5 V			GND				4.5 V		40		

1/ Delta computation shall be performed on parameters marked with an asterisk (*) with data read and recorded ("10X" denotes "10 times the initial measurement"). Where two delta values are indicated for one parameter, the greater value shall apply.
2/ Test sequence is optional with the manufacturer.
3/ Unless otherwise specified, the limits apply at T_A = -55°C to +125°C.
4/ A = Normal clock pulse
B = Momentary GND, then 4.5 V
C = Momentary GND, then open
D = Momentary 4.5 V, then GND
5/ Test time limits ≤ 100 ms
6/ Unused input pins may be connected to 4.5 V.

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TABLE I. (Device type 05) Electrical characteristics tests.

Test No. 1/2/	Characteristics	Symbol	Conditions	Limits 3/		Delta (Δ) 1/	Units
				Min	Max		
1	Input voltage required to ensure a high level input (each input)	V_{IH}	V_{CC} = 4.5 V	2			V
2	Input voltage required to ensure a low level input (each input)	V_{IL}	V_{CC} = 4.5 V		0.8		V
3*	High level output voltage	V_{OH}	V_{CC} = 4.5 V, V_{IH} = 2 V V_{IL} = 0.8, I_{OH} = -400 µA	2.4		±.12	V
4*	Low level output voltage	V_{OL}	V_{CC} = 4.5 V, V_{IH} = 2 V V_{IL} = 0.8 V, I_{OL} = 16 mA		0.4	±.03	V
5	Low level input current at preset or D	I_{IL1}	V_{CC} = 5.5 V, V_{IL} = 0.4 V	-.7	-1.6		mA
6	Low level input current at clear or clock	I_{IL2}	V_{CC} = 5.5 V, V_{IL} = 0.4 V	-1.4	-3.2		mA
7*	High level input current at D	I_{IH1}	V_{CC} = 5.5 V, V_{IH} = 2.4 V		40	10X or 4 µA	µA
		I_{IH2}	V_{CC} = 5.5 V, V_{IH} = 5.5 V		100		µA
8	High level input current at preset or clock	I_{IH3}	V_{CC} = 5.5 V, V_{IH} = 2.4 V		80		µA
		I_{IH4}	V_{CC} = 5.5 V, V_{IH} = 5.5 V		200		µA
9	High level input current at clear	I_{IH5}	V_{CC} = 5.5 V, V_{IH} = 2.4 V		120		µA
		I_{IH6}	V_{CC} = 5.5 V, V_{IH} = 5.5 V		300		µA
10	Short circuit output current	I_{OS}	V_{CC} = 5.5 V, V_{IL} = 0 V	-20	-57		mA
11	Supply current	I_{CC}	V_{CC} = 5.5 V, V_{IH} = 4.5 V		30		mA
12	Input clamp voltage	V_{IC}	V_{CC} = 4.5 V I_{IN} = -12 mA T_A = 25°C		-1.5		V

1/ Delta computation shall be performed on parameters marked with an asterisk (*) with data read and recorded ("10X" denotes "10 times the initial measurement"). Where two delta values are indicated for one parameter, the greater value shall apply.
2/ Test sequence is optional with the manufacturer.
3/ Unless otherwise specified, the limits apply at T_A = -55°C to +125°C.

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TABLE II. (Device type 01) Switching characteristics.

Characteristic	Symbol	Conditions	Test figure	Temp.	Limits Min	Limits Max	Units
Maximum clock frequency	f_{MAX}	V_{CC} = 5V, C_L = 50 pF, R_L = 390 ohms ±5%	5	25°C	5	---	MHz
				125°C	5	---	
				-55°C	5	---	
Propagation delay time to high level from clear or preset to output	t_{PLH1}		4	25°C	5	25	ns
				125°C	5	39	
				-55°C	5	39	
Propagation delay time to low level from clear or preset to output	t_{PHL_1}		4	25°C	5	40	ns
				125°C	5	50	
				-55°C	5	50	
Propagation delay time to high level from clock to output	t_{PLH_2}		5	25°C	5	30	ns
				125°C	5	39	
				-55°C	5	39	
Propagation delay time to low level from clock to output	t_{PHL_2}		5	25°C	5	40	ns
				125°C	5	50	
				-55°C	5	50	

TABLE II. (Device type 02) Switching characteristics.

Characteristic	Symbol	Conditions	Test figure	Temp.	Limits Min	Limits Max	Units
Maximum clock frequency	f_{MAX}	V_{CC} = 5V, C_L = 50 pF, R_L = 390 ohms ±5%	7	25°C	5	---	MHz
				125°C	5	---	
				-55°C	5	---	
Propagation delay time to high level from clear to output	t_{PLH_1}		6	25°C	5	25	ns
				125°C	5	39	
				-55°C	5	39	
Propagation delay time to low level from clear to output	t_{PHL_1}		6	25°C	5	40	ns
				125°C	5	50	
				-55°C	5	50	
Propagation delay time to high level from clock to output	t_{PLH_2}		7	25°C	5	30	ns
				125°C	5	39	
				-55°C	5	39	
Propagation delay time to level from clock to output	t_{PHL_2}		7	25°C	5	40	ns
				125°C	5	50	
				-55°C	5	50	

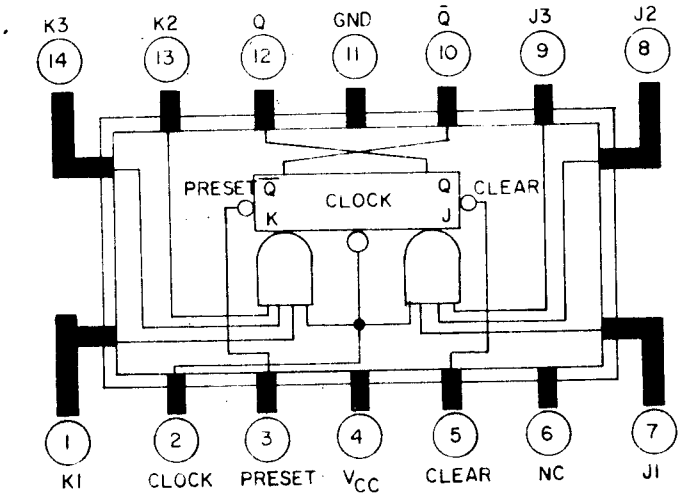
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TABLE II. (Device type 05) Switching characteristics.

Characteristic	Symbol	Conditions	Test figure	Temp.	Limits Min	Limits Max	Units
Maximum clock frequency	f_{MAX}	V_{CC} = 5V C_L = 50 pF R_L = 390 ohms ±5%	9	25°C	5	---	MHz
				125°C	5	---	
				-55°C	5	---	
Minimum input setup time	t_{setup}			25°C	---	20	ns
Minimum input hold time	t_{hold}			25°C	---	5	ns
Propagation delay time to high level from clear or preset to output	t_{PLH_1}		8	25°C	5	25	ns
				125°C	5	39	
				-55°C	5	39	
Propagation delay time to low level from clear or preset to output	t_{PHL_1}		8	25°C	5	40	ns
				125°C	5	50	
				-55°C	5	50	
Propagation delay time to high level from clock to output	t_{PLH_2}		9	25°C	5	30	ns
				125°C	5	39	ns
				-55°C	5	39	ns
Propagation delay time to low level from clock to output	t_{PHL_2}		9	25°C	5	40	ns
				125°C	5	50	ns
				-55°C	5	50	ns

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Device type 01



Device type 02

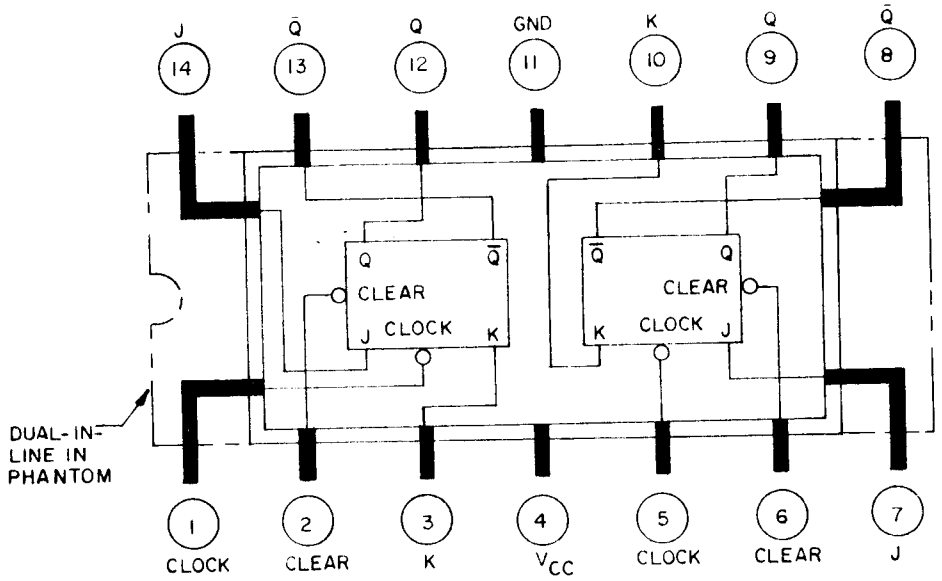


FIGURE 1. Logic diagrams and terminal connections.

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Device type 05

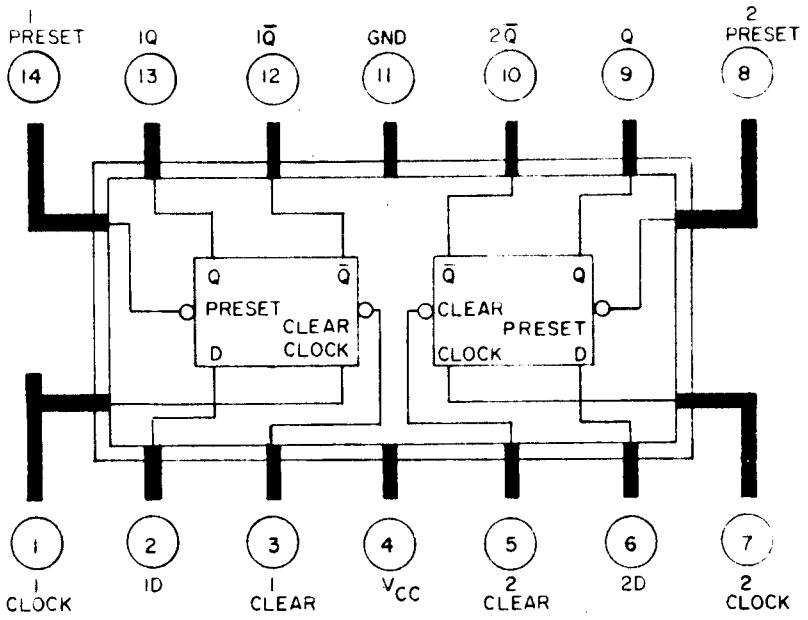


FIGURE 1. Logic diagrams and terminal connections - Continued.

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Device type 01

Truth table		
t_n		t_{n+1}
J	K	Q
L	L	Q_n
L	H	L
H	L	H
H	H	$\overline{Q}_n$

Positive logic: Low input to preset sets Q to high-level.
Low input to clear sets Q to low-level.
Preset and clear are independent of clock and dominate regardless of the state of clock or J or K inputs.

NOTES:
1. J = J1 · J2 · J3
2. K = K1 · K2 · K3
3. t_n = Bit time before clock pulse.
4. t_{n+1} = Bit time after clock pulse.

Device type 02

Truth table each flip-flop		
t_n		t_{n+1}
J	K	Q
L	L	Q_n
L	H	L
H	L	H
H	H	$\overline{Q}_n$

Positive logic: Low input to clear sets Q to low-level.
Clear is independent of clock and dominate regardless of the state of clock or J or K inputs.

NOTES: 1. t_n = Bit time before clock pulse.
2. t_{n+1} = Bit time after clock pulse.

Device type 05

Truth table each flip-flop		
t_n		t_{n+1}
INPUT D	OUTPUT Q	OUTPUT $\overline{Q}$
L	L	H
H	H	L

Positive logic:

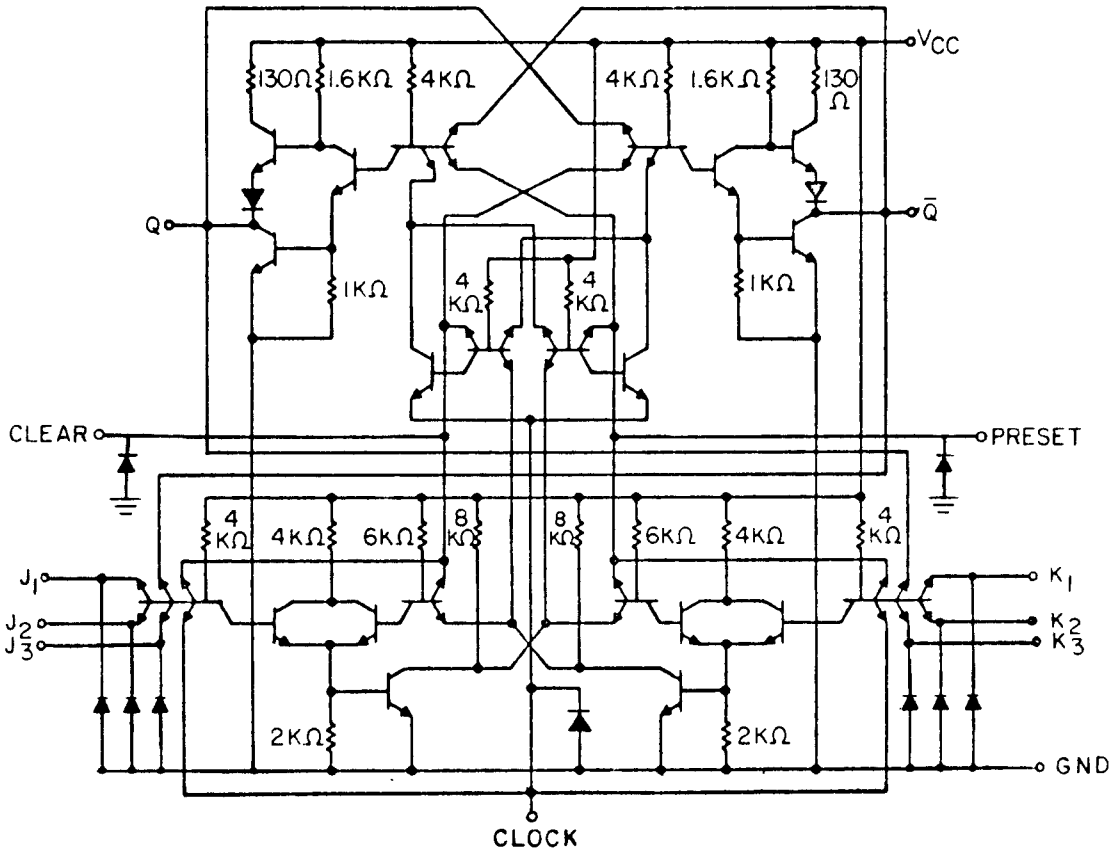
Low input to preset sets Q to high-level.
Low input to clear sets Q to low-level.
Clear and preset are independent of clock and dominate regardless of the state of clock or D input.

NOTES: 1. t_n = Bit time before clock pulse.
2. t_{n+1} = Bit time after clock pulse.

FIGURE 2. Truth tables and logic equations.

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Device type 01

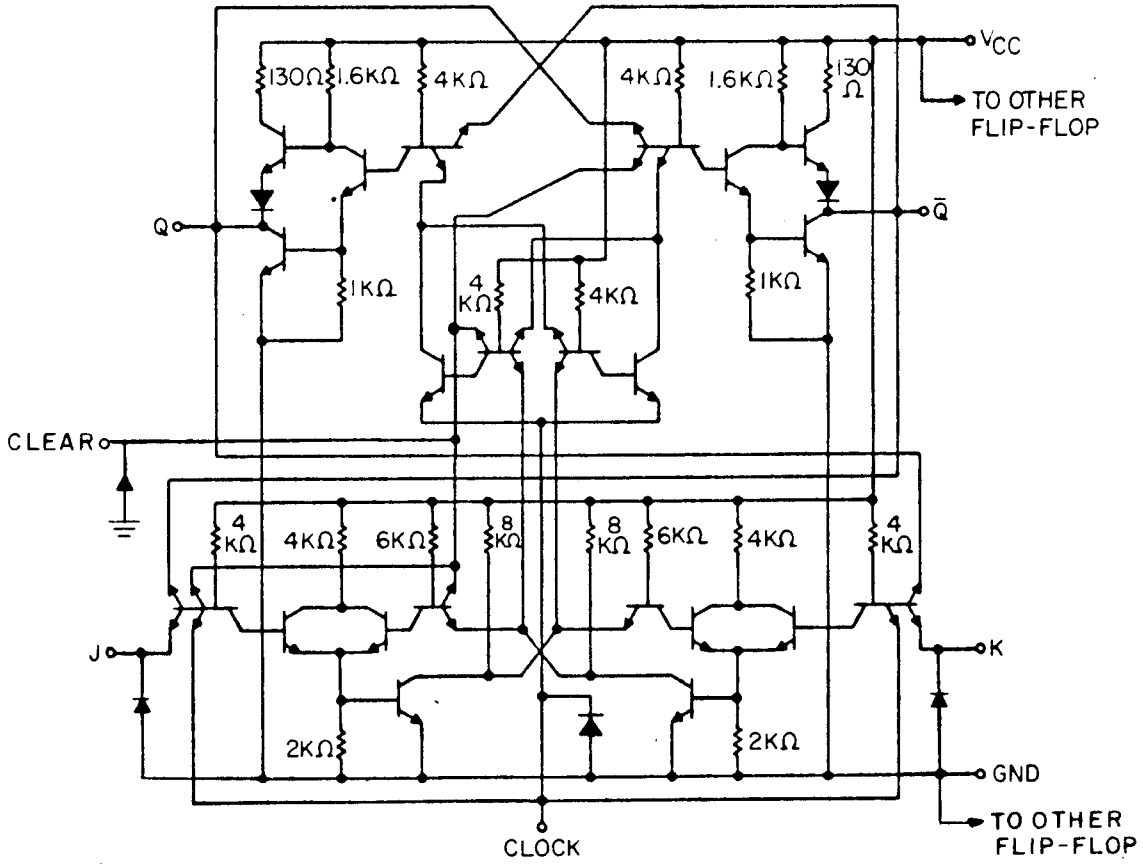


NOTE: All resistor values shown are nominal.

FIGURE 3. Schematic circuits.

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Device type 02



NOTE: Resistor values shown are nominal.

FIGURE 3. Schematic circuits - Continued.

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Device type 05

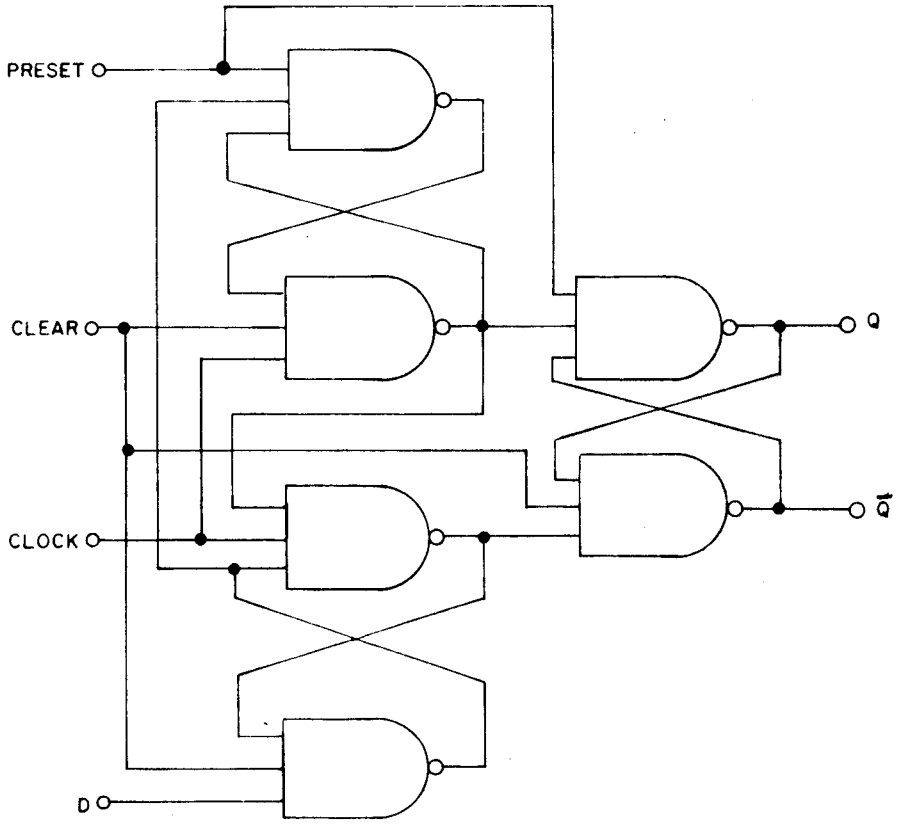
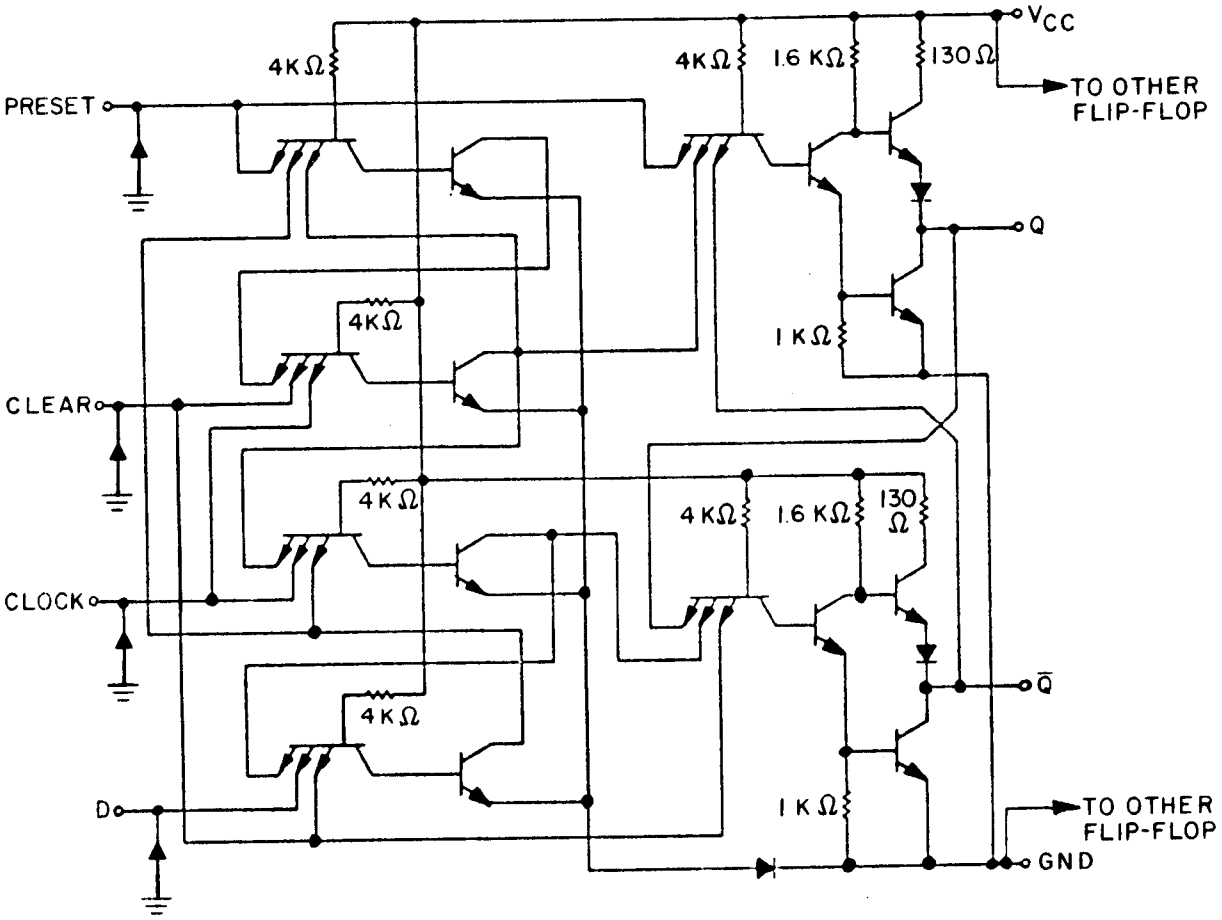


FIGURE 3. Schematic circuits - Continued.

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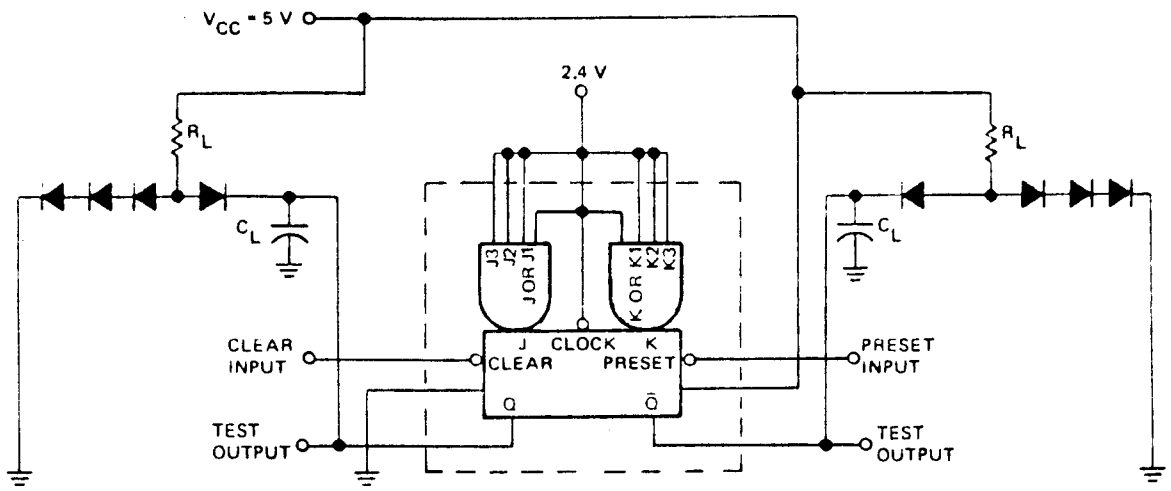
Device type 05



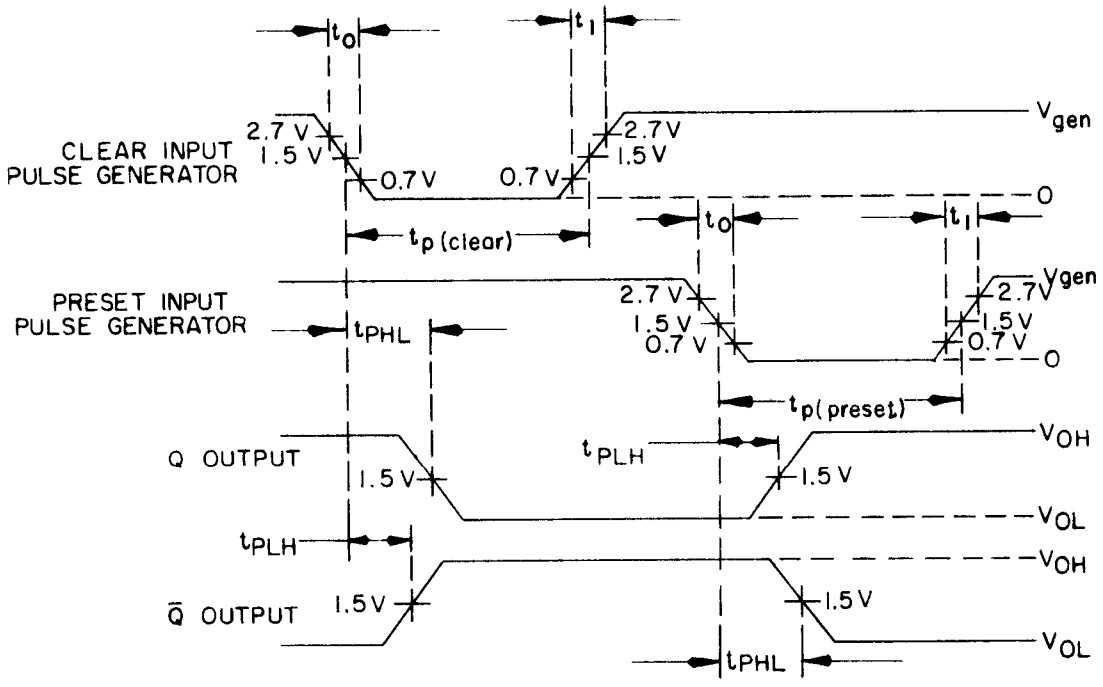
NOTE: All resistance values shown are nominal.

FIGURE 3. Schematic circuits - Continued.

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TEST CIRCUIT

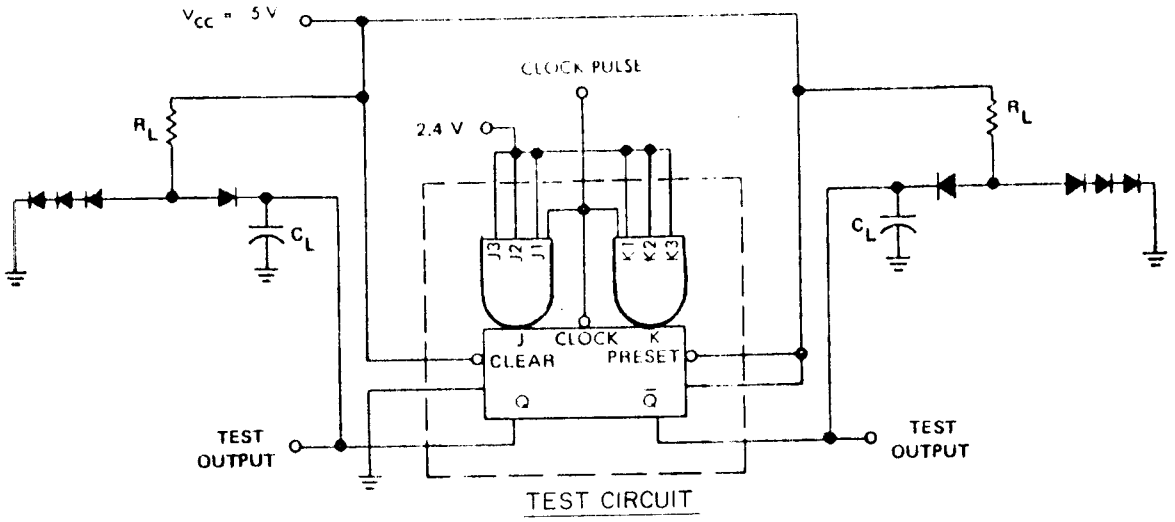


VOLTAGE WAVEFORMS

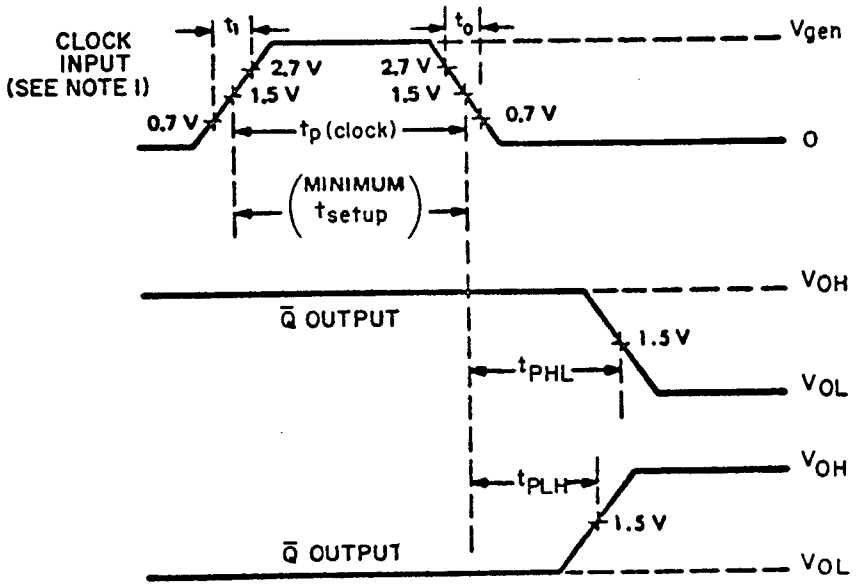
NOTES:
1. Clear or preset inputs dominate regardless of the state of clock or J-K inputs.
2. Clear or preset input pulse characteristics: V_{gen} = 3.0 V, $t_0 = t_1 \leq 7$ ns, t_p(clear) = t_p(preset) = 25 ns, PRR = 1 MHz, and $Z_{out} \approx 50\Omega$.
3. C_L = 50 pF ±5%, including jig and probe capacitance.
4. R_L = 390Ω ±5%.
5. All diodes are 1N3064, or equivalent.

FIGURE 4. Clear and preset switching test circuit and waveforms for device type 01.

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TEST CIRCUIT

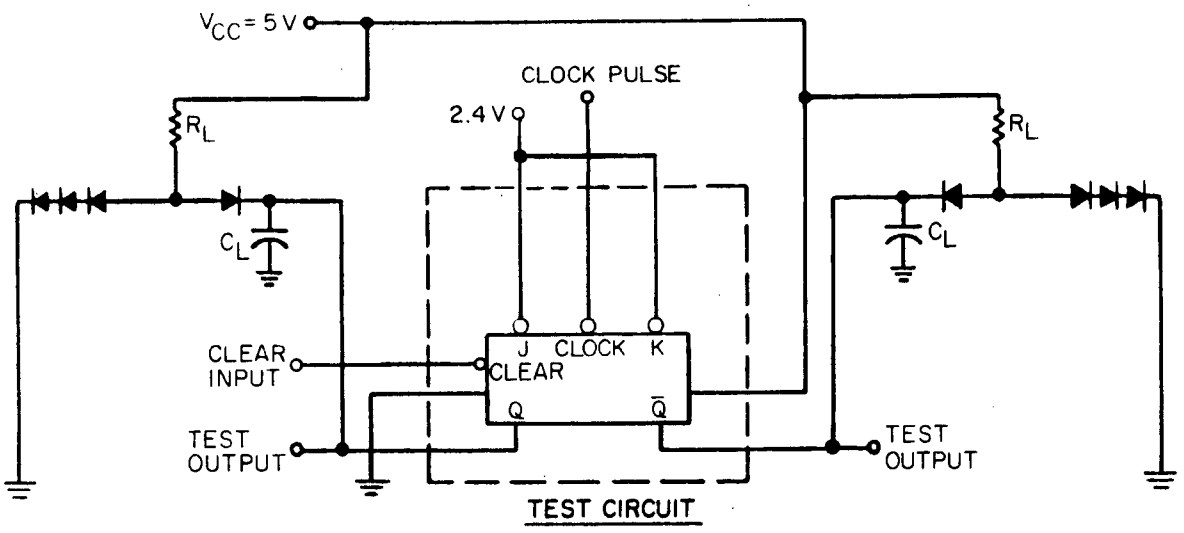


VOLTAGE WAVEFORMS

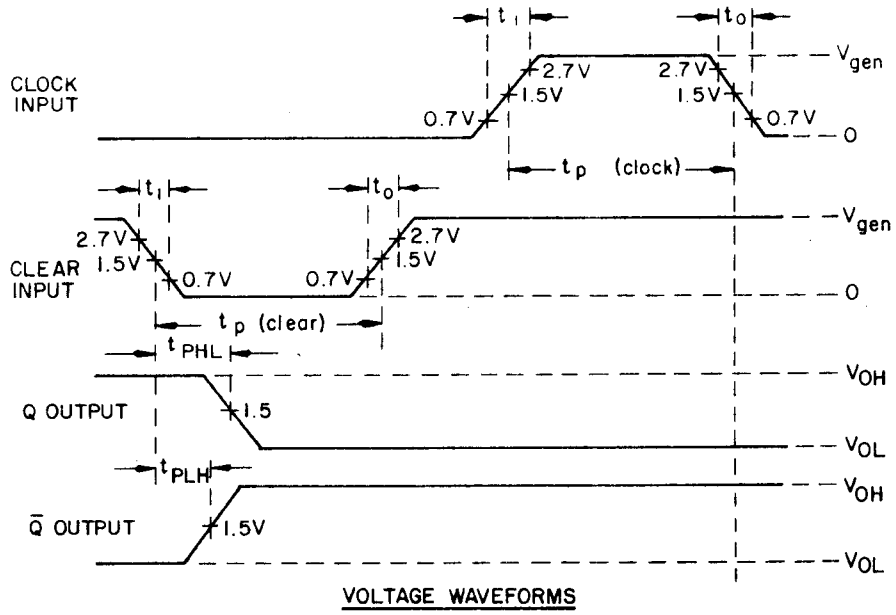
NOTES:
1. Clock input pulse characteristics: V_{gen} = 3.0 V, $t_0 = t_1 \leq 7$ ns, t_p = 25 ns, and PRR = 1 MHz. When testing F_{MAX}, PRR = 10 MHz.
2. J = J1 · J2 · J3; and K = K1 · K2 · K3.
3. All diodes are 1N3064, or equivalent.
4. C_L = 50 pF ±5%, including jig and probe capacitance.
5. R_L = 390Ω ±5%.
6. The output Q will be established by the J and K data which is available at the positive edge of the clock pulse.

FIGURE 5. Synchronous switching test circuit for device type 01.

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TEST CIRCUIT

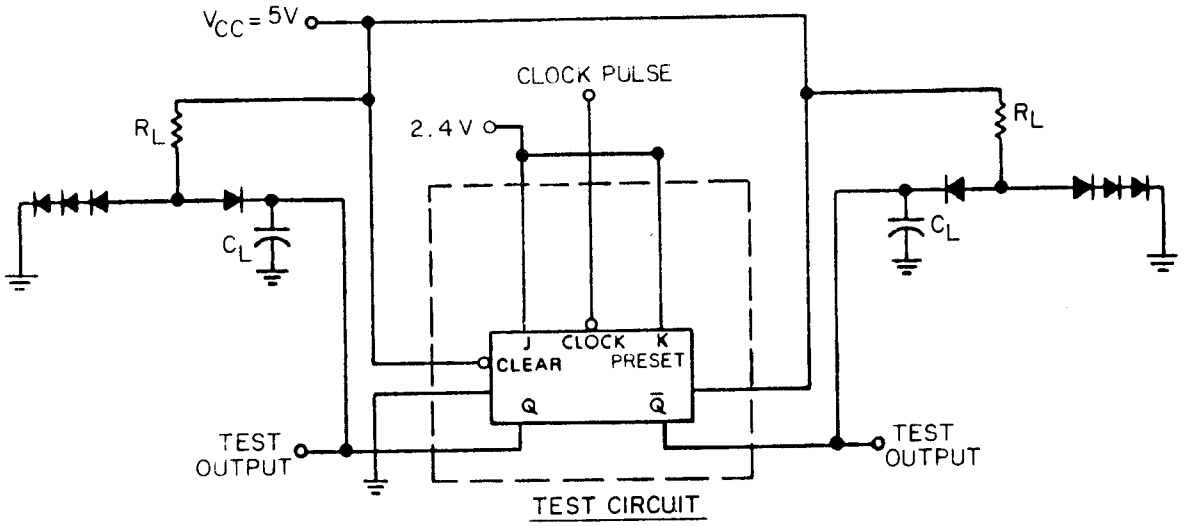


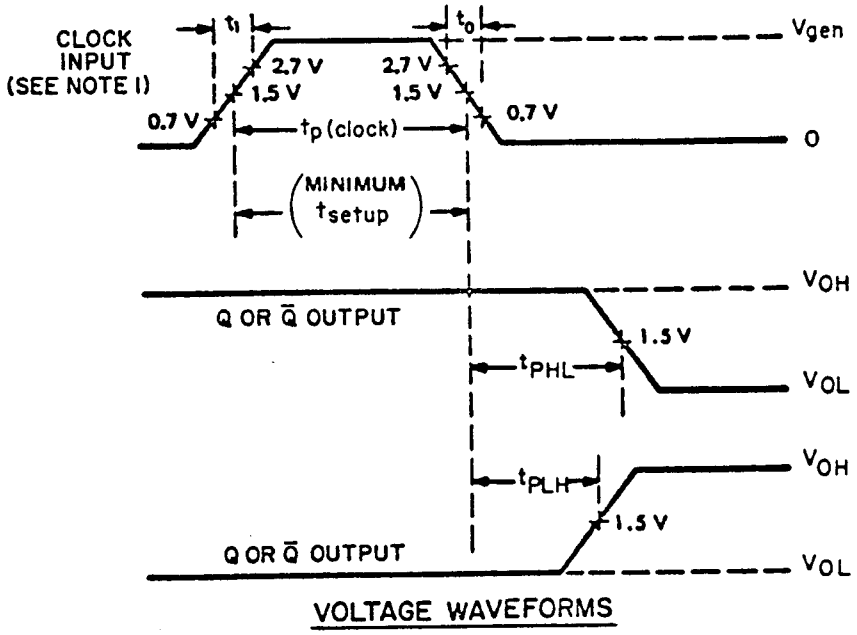
VOLTAGE WAVEFORMS

NOTES:
1. Clear input dominate regardless of the state of clock or J-K inputs.
2. Clear input pulse characteristics: V_{gen} = 3.0 V, $t_0 = t_1 \leq$ 7 ns, t_p(clear) = 25 ns, PRR = 1 MHz.
3. All diodes are 1N3064, or equivalent.
4. C_L = 50 ±5%, including jig and probe capacitance.
5. R_L = 390Ω ±5%.
6. Clock input pulse characteristics: V_{gen} = 3.0 V, $t_0 = t_1 \leq$ 7 ns, t_p(clock) = 25 ns, PRR = 1 MHz.

FIGURE 6. Clear switching test circuit and waveforms for device type 02.

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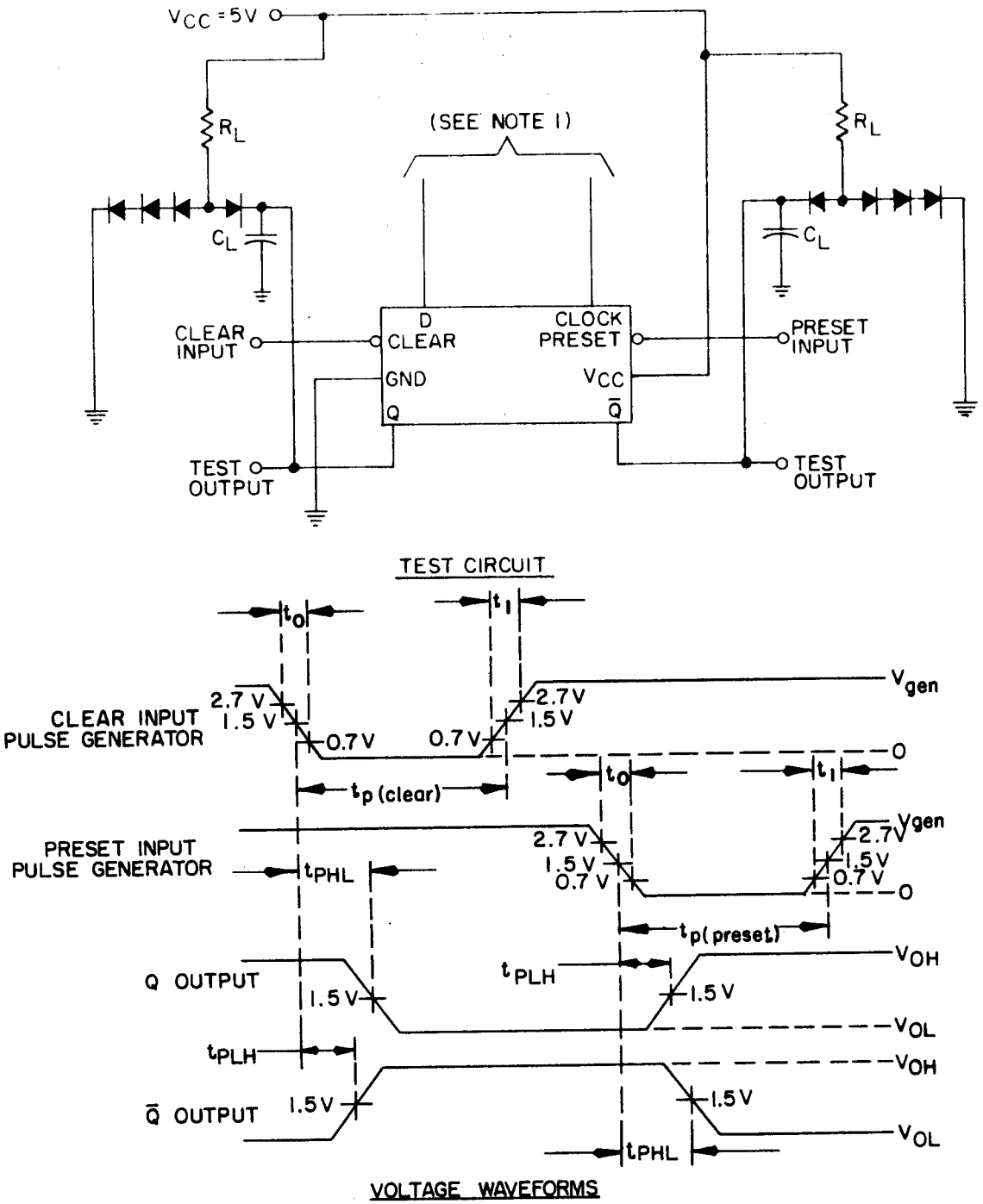




NOTES:
1. Clock J and K input pulse characteristics: V_{gen} = 3.0 V, $t_0 = t_1 \leq$ 7 ns, t_p = 25 ns, and PRR = 1 MHz. When testing F_{MAX}, PRR = 15 MHz.
2. All diodes are 1N3064.
3. C_L = 50 pF ±5%, including jig and probe capacitance.
4. R_L = 390Ω ±5%.

FIGURE 7. Synchronous switching test circuit for device type 02.

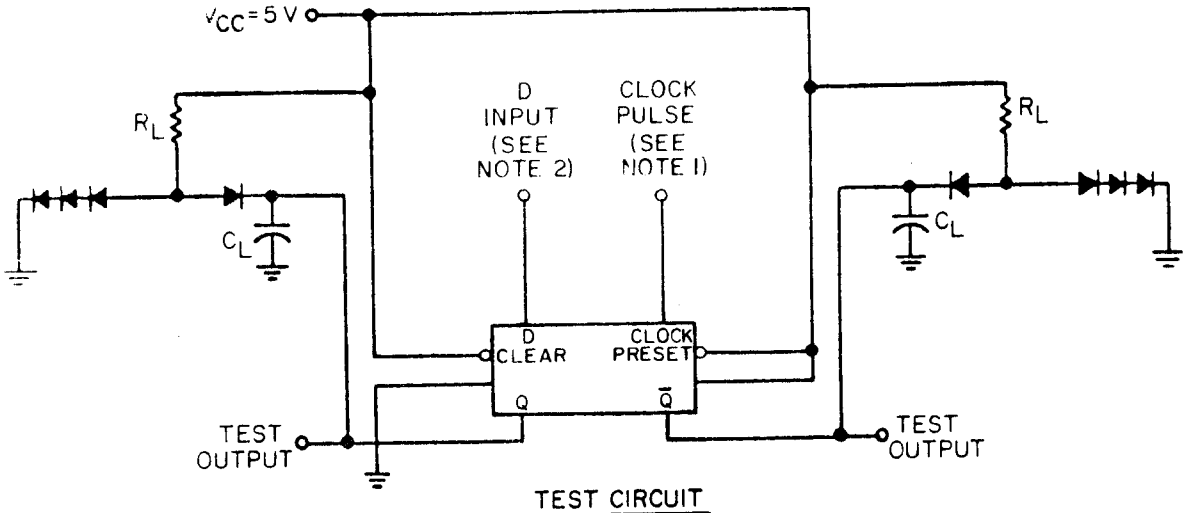
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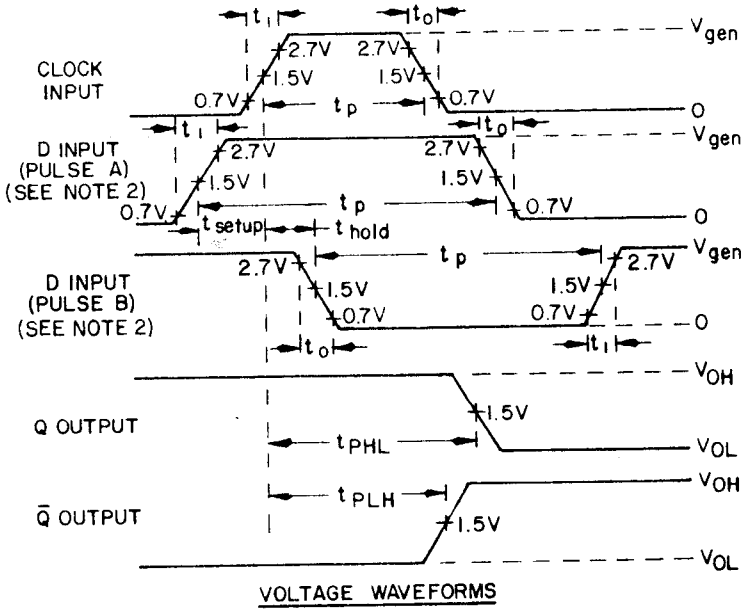
NOTES:
1. Clear and preset inputs dominate regardless of the state of clock or D inputs.
2. All diodes are 1N3064, or equivalent.
3. Clear or preset input pulse characteristics: V_{gen} = 3.0 V, $t_0 = t_1 \leq 7$ ns, t_p = 30 ns and PRR = 1 MHz.
4. C_L = 50 pF ±5%, including jig and probe capacitance.
5. R_L = 390Ω ±5%.

FIGURE 8. Clear and preset switching test circuit and waveforms for device type 05.

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TEST CIRCUIT



VOLTAGE WAVEFORMS

NOTES:
1. Clock input pulse has the following characteristics: V_{gen} = 3.0 V, $t_0 = t_1 \leq 7$ ns, t_p = 30 ns, and PRR = 1 MHz. When testing F_{MAX}, PRR = 10 MHz.
2. D input (pulse A) has the following characteristics: V_{gen} = 3.0 V, $t_0 = t_1 \leq 7$ ns, t_{setup} = 20 ns, t_p = 60 ns, and PRR is 50 percent of the clock PRR. D input (pulse B) has the following characteristics: V_{gen} = 3.0 V, $t_0 = t_1 \leq 7$ ns, t_{hold} = 5 ns, t_p = 60 ns, and PRR is 50 percent of the clock PRR. Propagation delay times shall be measured using each input. When F_{MAX}, D input is connected to Q.
3. All diodes are 1N3064 or equivalent.
4. C_L = 50 pF ±5%, including jig and probe capacitance.
5. R_L = 390Ω ±5%.

FIGURE 9. Synchronous switching test circuit for device type 05.

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3.6 Extended life tests. Extended life tests shall be performed in accordance with MIL-M-0038510.

3.7 Marking. Marking shall be in accordance with MIL-M-0038510 and 1.2 herein.

4. PRODUCT ASSURANCE PROVISIONS

4.1 Product assurance provisions. Product assurance shall be in accordance with MIL-M-0038510.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery. Preparation for delivery shall be as specified in MIL-M-0038510.

6. NOTES

6.1 Notes. Notes are as specified in MIL-M-0038510.

Custodian:
Air Force - 19

Review activities:
Air Force - 17, 19
DSA - ES

Preparing activity:
Air Force - 19

Agent:
DSA - ES

(Project 5962-F088-50)

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